



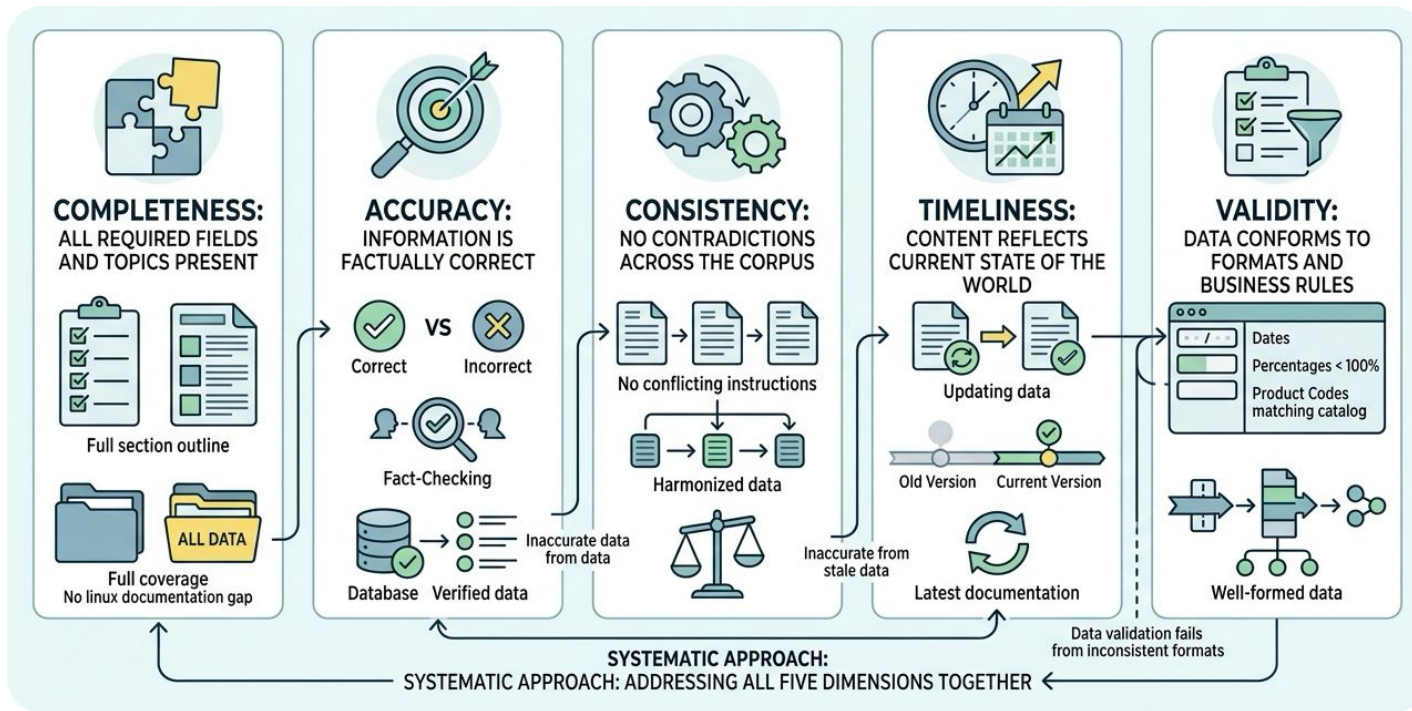
U.S. General Services
Administration



Chapter 6.4: Data Quality Fundamentals

US General Service Administration
AI Community of Practice - March 2026

Five Dimensions of Data Quality



Completeness -- The Three Levels

FIELD-LEVEL COMPLETENESS



- MISSING VALUES IN STRUCTURED RECORDS
- FIELD GAPS ARE DETECTABLE (e.g., Schema Validation)



DOCUMENT-LEVEL COMPLETENESS



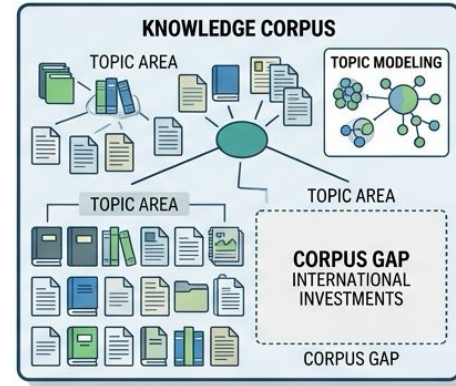
DIAGNOSTIC
STEPS



RESOLUTION
PROCEDURES

- ARTICLES MISSING KEY SECTIONS
- TRIGGERS USER "I DON'T KNOW" MESSAGES OR UNSURE AGENTS

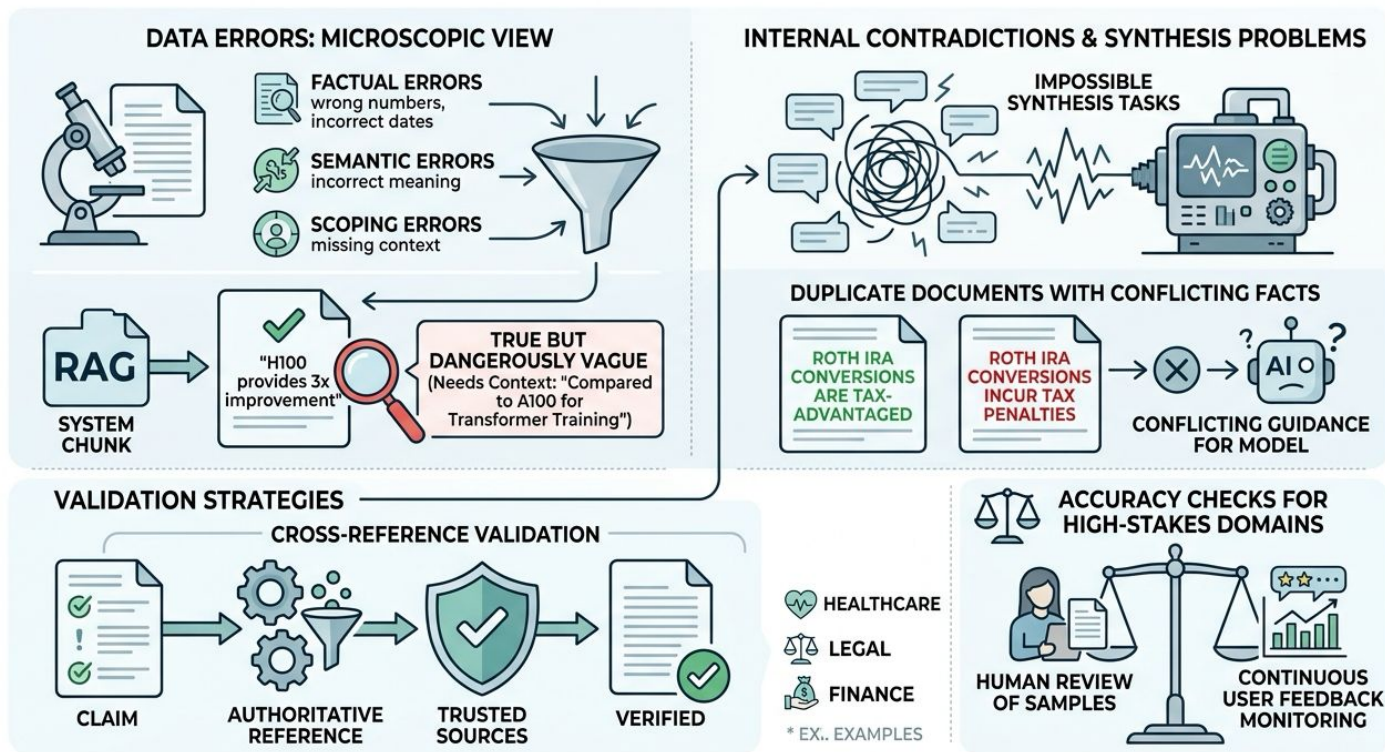
CORPUS-LEVEL COMPLETENESS



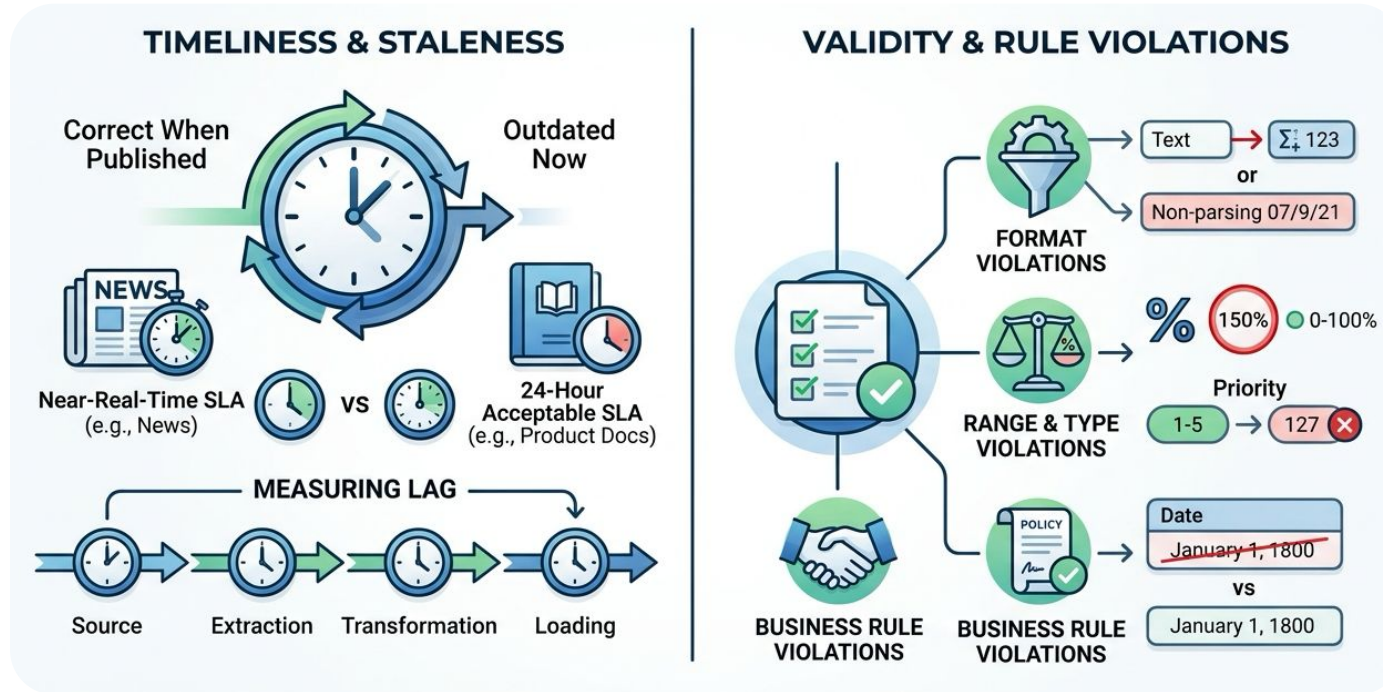
- ENTIRE TOPIC AREAS UNCOVERED
- ** CORPUS GAPS REQUIRE TOPIC MODELING (to map against required taxonomy)

INCOMPLETE DATA FORCES AGENTS TO GUESS OR SAY 'I DON'T KNOW'

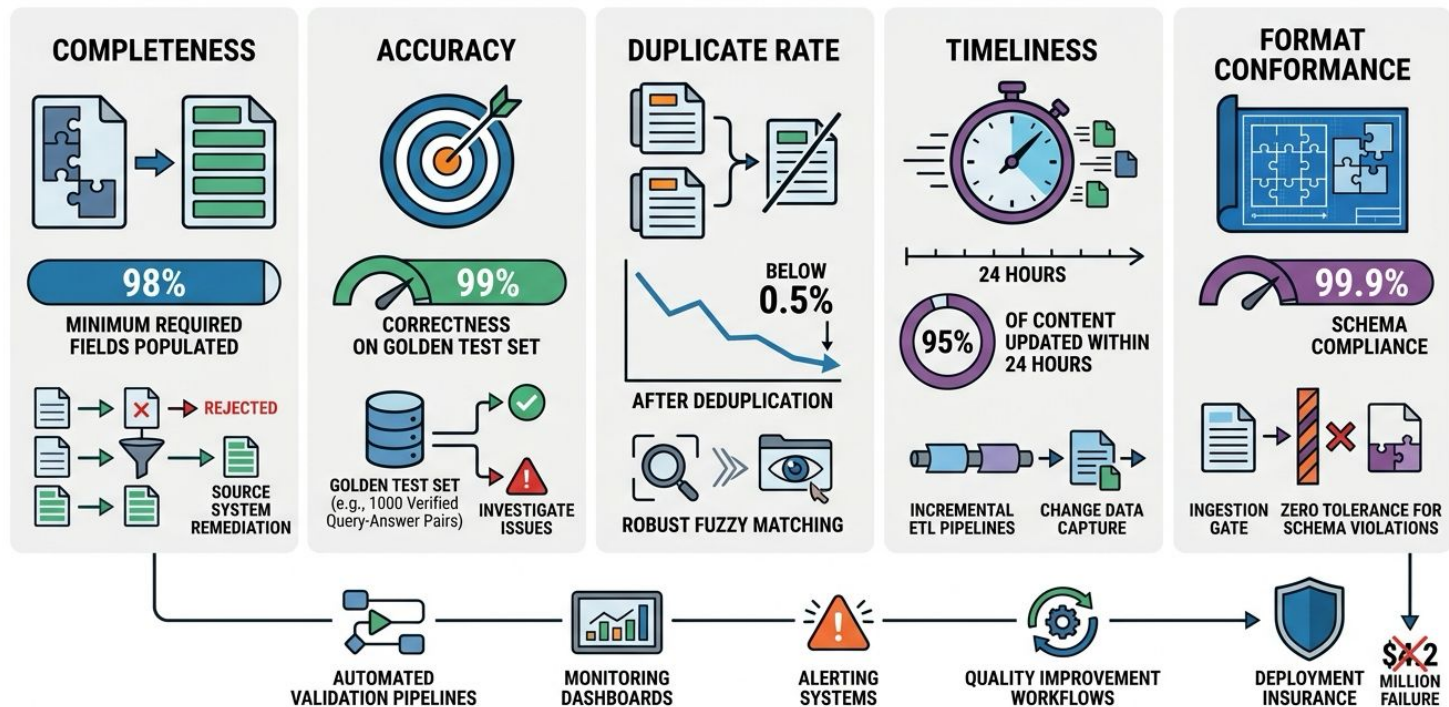
Accuracy and Consistency Under the Microscope



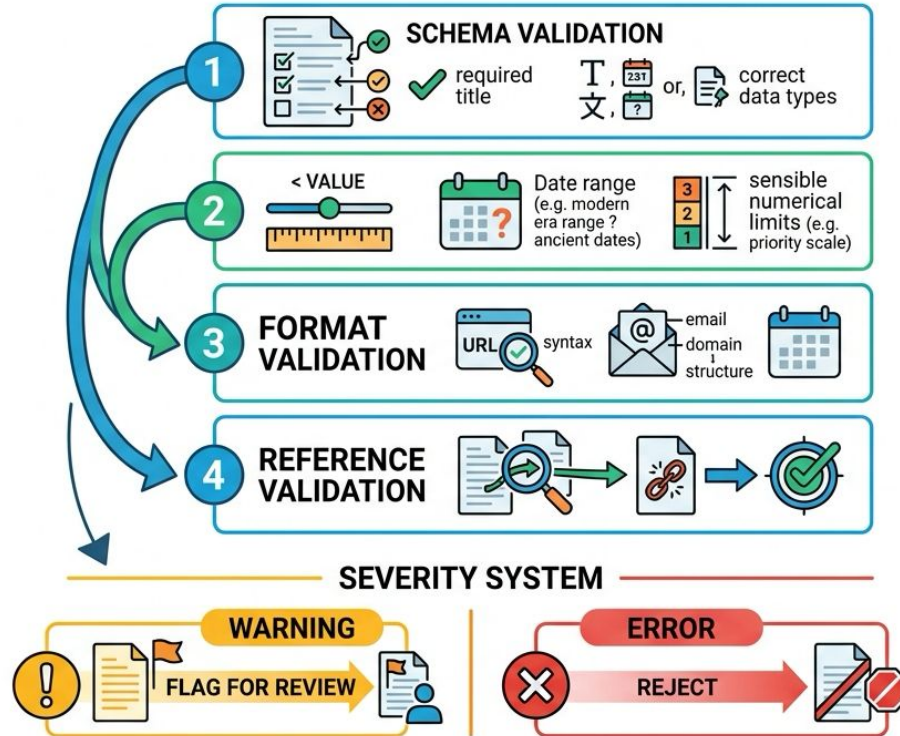
Timeliness and Validity -- The Quiet Killers



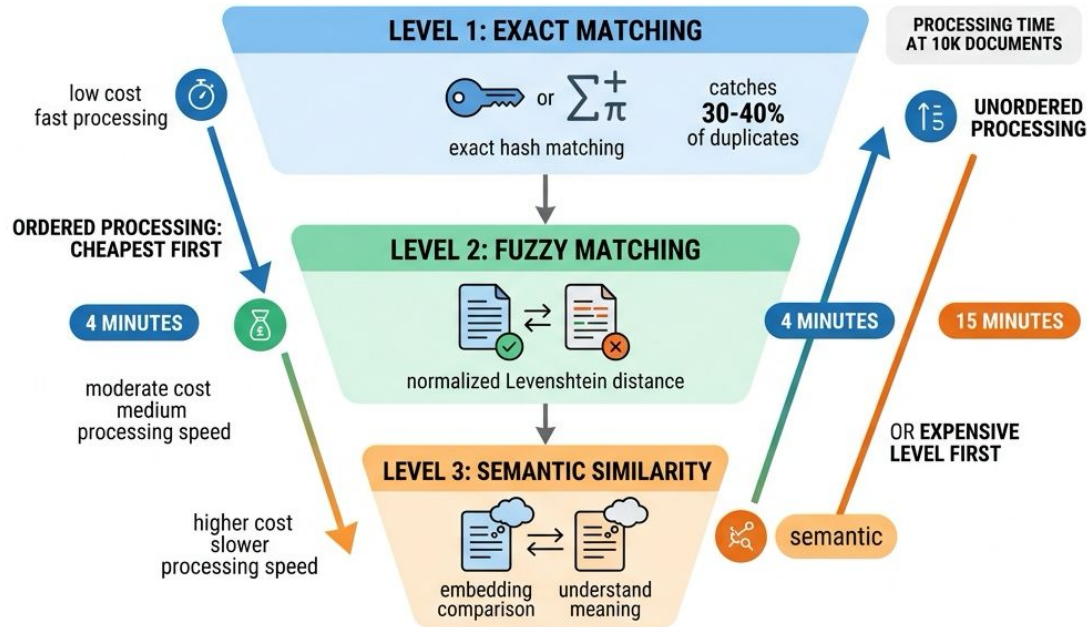
Quality SLAs -- Making It Measurable



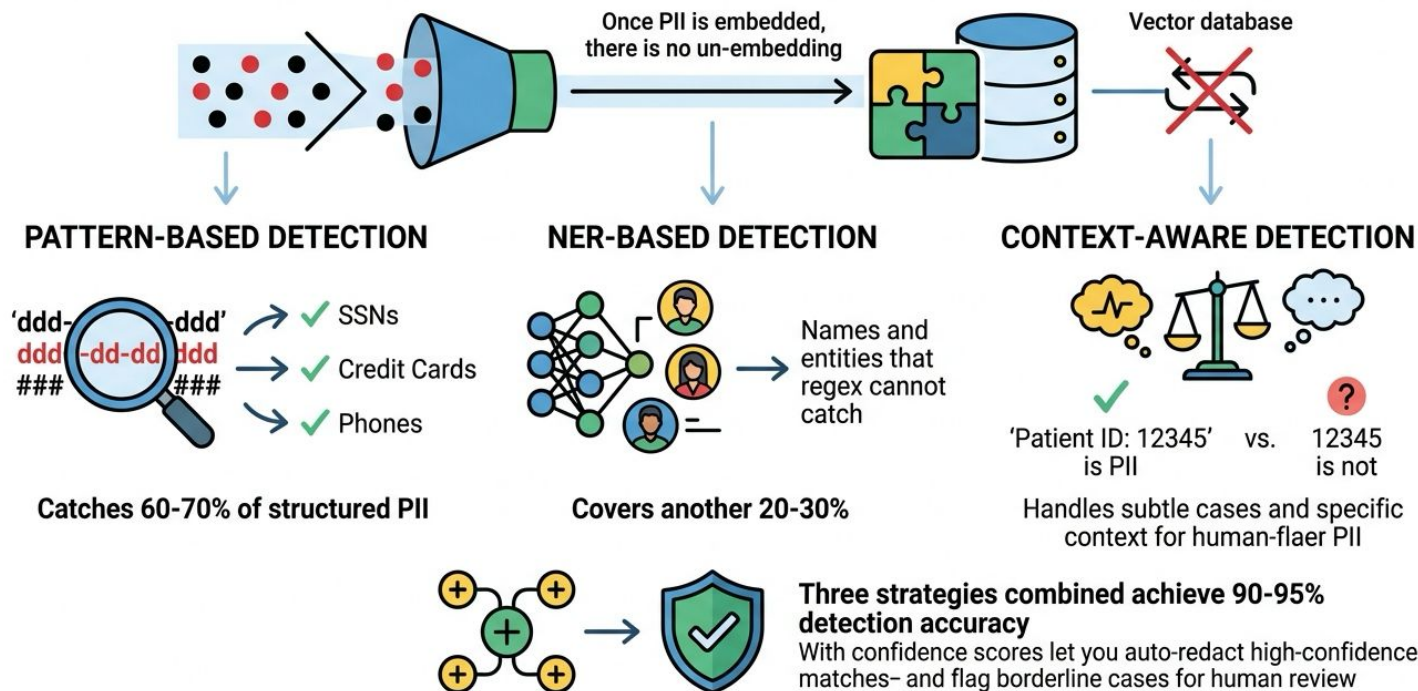
The Four-Level Validation Framework



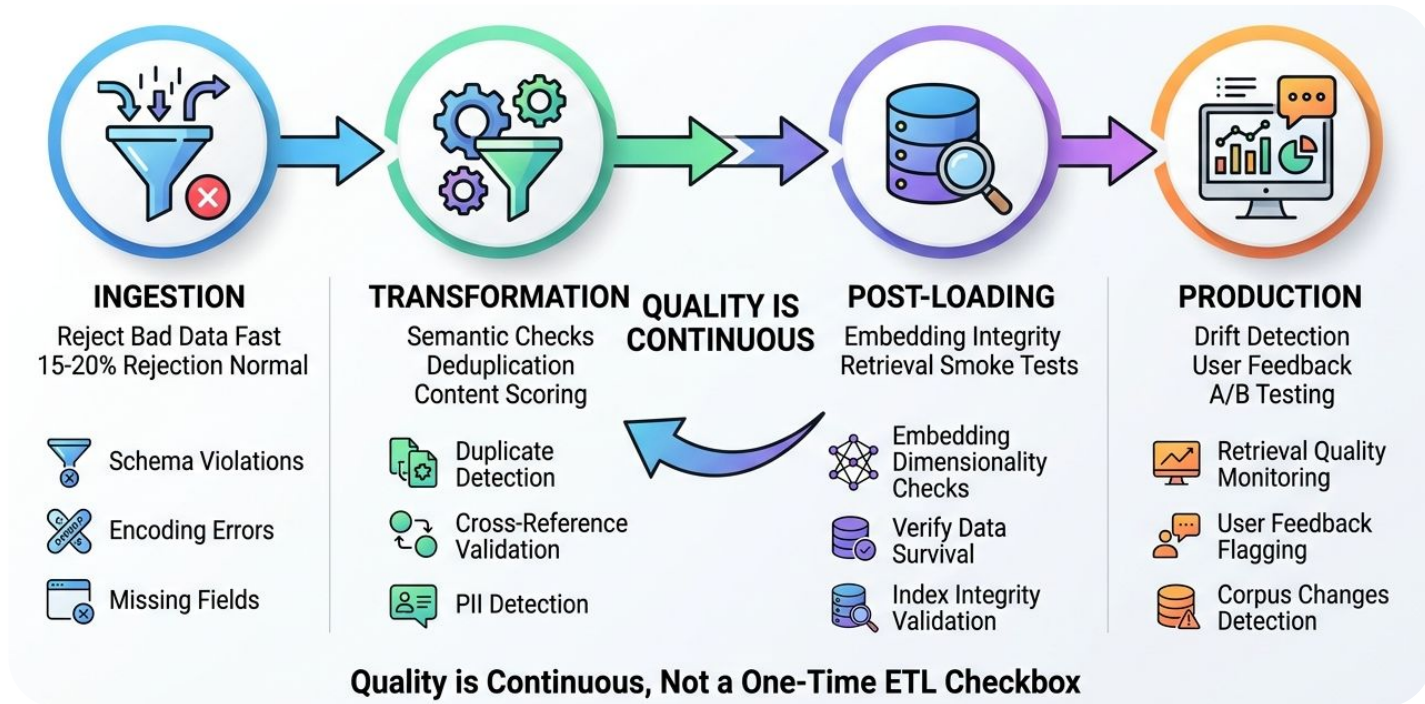
Three-Level Deduplication Strategy



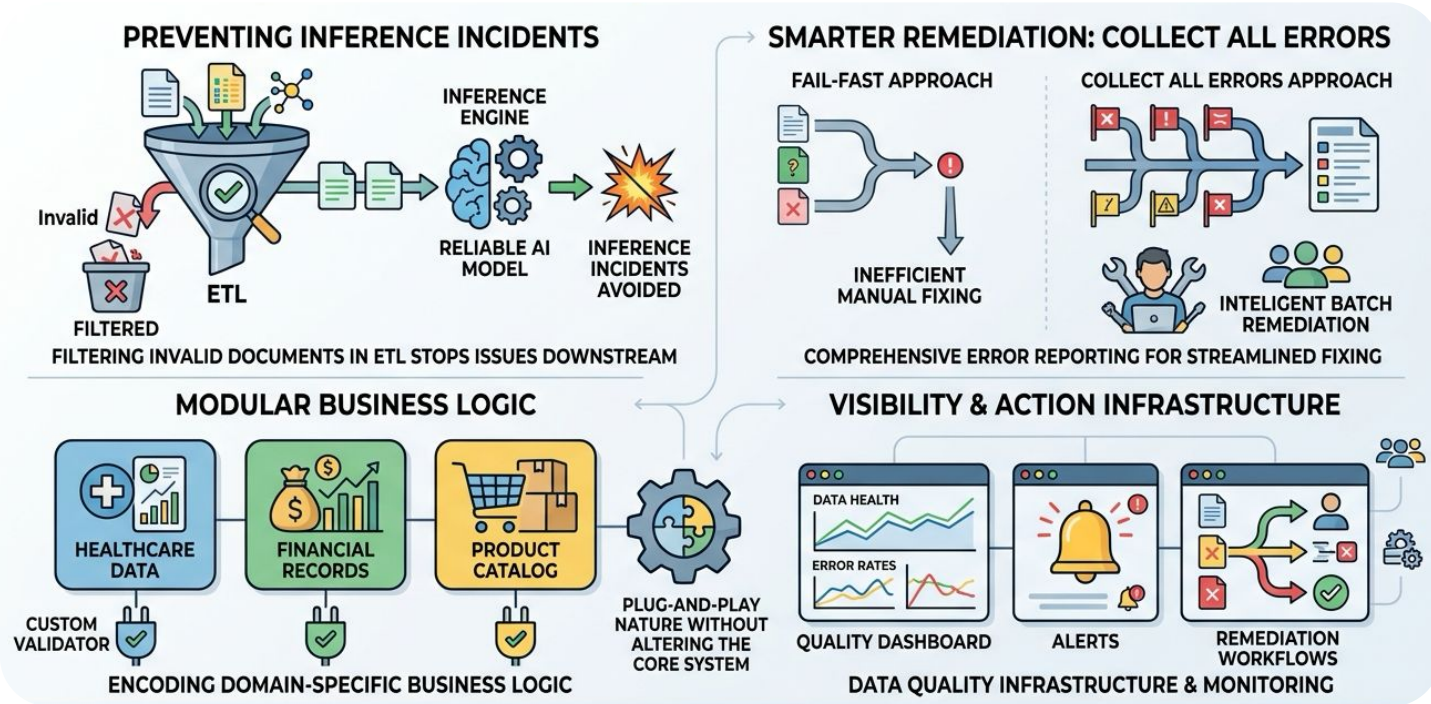
PII Detection Before It Is Too Late



Quality Gates Across the Agent Lifecycle



The Data Quality Investment



Key Takeaways

- Data quality is the foundation, not a feature, of reliable RAG
 - Five dimensions: completeness, accuracy, consistency, timeliness, validity
 - Validate at four levels: schema, range, format, reference
 - Deduplicate in three tiers; detect PII before embedding
 - Next up: Chapter 6.5 brings production RAG architecture together
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